

1 · Data

Public dataset

Nature 2025 · MIT licence

341 formulations

6 monomers on the
composition simplex

Target: adhesion
strength (kPa)

2 · Training region

Training set

n = 316 · full adhesion range

5-fold grouped CV

5 seeds · 25 models

Ablation-gated
components

3 · SIMPLEX

SIMPLEX

dual-modality encoder

Monomers +
pairwise terms

ResBlock x2 +
interaction attention

Mixup · SWA ·
domain constraint

4 · Prospective validation

Prospective cohort

n = 25 · final SMBO

High-performance
composition region

Held-out cohort
(62–251 kPa)

Evaluated once,
after freezing

5 · Screening

Ranking + insight

Spearman ρ = 0.50
vs RF 0.21

Top-20 precision 0.25
vs RF 0.10

Permutation importance
→ composition synergy

Accelerated screening